

LOWE AHP, AL, USA

WMO: 722239

Lat: 31.356N

Lon: 85.751W

Elev: 74

StdP: 100.43

Time Zone: -6.00 (NAC)

Period: 08-19

WBAN: 53861

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.9	-2.2	-12.1	1.3	2.5	-9.5	1.7	3.6	8.6	13.1	7.8	11.8	1.3	330	0.342

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	35.8	23.9	34.6	24.1	33.6	24.1	26.3	31.8	25.8	31.2	25.5	30.8	2.6	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.0	20.2	28.3	24.5	19.6	27.9	24.0	19.1	27.6	81.8	32.0	80.0	31.3	78.3	31.0	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.2	5.8	5.0	-6.9	36.7	2.2	1.7	-8.4	37.9	-9.7	38.9	-11.0	39.8	-12.5	41.0		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			-7.8	27.2	2.3	0.5	-9.4	27.5	-10.8	27.8	-12.1	28.0	-13.8	28.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.2	9.4	12.2	15.1	18.8	23.0	26.7	27.6	27.2	25.2	19.6	13.7	11.7
	DBStd	7.52	5.90	5.24	4.82	3.23	3.04	1.80	1.50	1.60	2.47	4.24	4.68	5.27
	HDD10.0	194	86	35	12	0	0	0	0	0	1	17	44	
	HDD18.3	1028	284	178	124	32	6	0	0	1	37	152	215	
	CDD10.0	3562	66	97	169	266	404	501	546	534	457	298	128	96
	CDD18.3	1355	6	7	23	48	152	251	288	276	207	77	13	8
	CDH23.3	12883	15	38	194	515	1554	2492	2828	2611	1811	703	90	31
	CDH26.7	5641	0	2	29	113	652	1226	1371	1209	811	217	11	1
Wind	WSAvg	1.7	2.1	2.2	2.3	2.0	1.6	1.6	1.3	1.3	1.3	1.5	1.6	1.8
Precipitation	PrecAvg	1467	145	122	149	117	76	142	167	121	109	82	114	92
	PrecMax	1909	357	242	301	224	232	282	574	218	409	261	316	153
	PrecMin	944	49	40	9	26	0	65	68	46	18	12	37	37
	PrecStd	276	74	61	77	59	50	74	110	49	89	59	70	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.2	26.7	29.1	30.7	35.1	37.6	36.7	36.7	36.1	32.5	28.2	25.9
		MCWB	18.9	20.2	18.2	20.5	22.1	22.8	25.0	23.5	23.3	21.6	21.0	20.6
	2%	DB	23.0	24.4	27.4	29.3	33.3	35.9	35.2	34.9	34.2	30.7	25.7	24.1
		MCWB	18.4	18.8	18.6	19.8	21.7	23.6	24.8	24.6	23.0	21.6	19.3	19.4
	5%	DB	21.0	22.4	25.7	27.9	31.7	34.1	34.1	33.7	32.7	29.2	24.0	22.4
		MCWB	17.9	17.7	17.8	18.9	21.3	23.9	24.8	24.8	23.3	20.9	18.4	18.8
	10%	DB	19.0	20.7	23.6	26.6	30.3	32.8	32.9	32.5	31.3	27.4	22.3	20.5
		MCWB	15.9	16.8	16.9	18.3	21.1	23.8	25.0	24.5	23.4	20.0	17.5	18.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.8	21.3	21.2	23.0	24.4	26.5	26.8	26.9	25.8	25.1	23.2	22.6
		MCDB	22.4	24.3	23.9	27.2	29.2	32.1	32.2	32.3	31.0	28.8	25.5	24.5
	2%	WB	19.7	20.3	20.3	22.1	23.6	25.7	26.3	26.3	25.2	24.0	21.7	20.9
		MCDB	21.8	23.0	23.6	25.8	28.6	31.3	32.0	31.6	30.3	27.0	23.3	22.6
	5%	WB	18.4	19.1	19.6	21.1	23.0	25.2	25.8	25.8	24.8	23.0	20.2	19.8
		MCDB	20.6	21.7	23.1	24.7	28.0	30.7	31.4	30.9	29.8	26.6	22.9	21.2
	10%	WB	16.8	17.8	18.5	20.0	22.5	24.7	25.5	25.5	24.3	22.0	18.5	18.3
		MCDB	19.1	20.0	22.7	23.9	27.6	30.0	31.0	30.4	29.2	25.2	21.6	20.2
Mean Daily Temperature Range	5% DB	MDBR	12.7	12.6	13.5	13.7	13.5	11.5	10.9	10.6	11.4	13.7	14.2	11.8
		MCDBR	12.1	12.9	14.1	15.2	15.3	14.0	12.3	12.4	13.5	14.2	14.0	12.0
		MCWBR	8.0	7.8	6.8	6.5	5.6	4.4	4.0	4.2	4.7	6.1	8.2	7.9
	5% WB	MCDBR	10.0	11.1	10.9	10.9	11.7	11.0	10.9	10.2	10.7	10.7	11.0	9.7
		MCWBR	7.5	7.8	6.2	6.2	4.9	4.3	4.2	4.0	4.4	5.3	7.4	7.6
Clear-Sky Solar Irradiance	taub	0.325	0.334	0.358	0.383	0.435	0.472	0.517	0.504	0.432	0.373	0.347	0.331	
	taud	2.520	2.496	2.431	2.364	2.253	2.209	2.095	2.127	2.309	2.448	2.468	2.519	
	Ebn at Noon	904	928	925	908	859	823	785	790	838	871	869	877	
	Edh at Noon	87	97	111	123	139	145	161	155	124	101	90	83	
All-Sky Solar Radiation	RadAvg	2.98	3.56	4.77	5.84	6.39	6.10	5.94	5.53	4.99	4.33	3.35	2.58	
	RadStd	0.19	0.45	0.40	0.48	0.44	0.48	0.37	0.36	0.38	0.48	0.28	0.28	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (46 neighbors)	+0.48	N/A	N/A	N/A	N/A	N/A	N/A	-93	+166	+109

Nomenclature: See separate page